

Lab 4

Aim : To implement and execute advanced SQL queries on a relational database using joins, subqueries, set operations, aggregate functions, grouping, and conditional operators for effective data retrieval and analysis.

Q1. Find match Ids of all the matches in the database in which tendulkar batted.

```
mysql> SELECT MatchID FROM Batting, Player;
```

```
mysql> SELECT MatchID FROM Batting, Player WHERE PID=PlayerID;
```

```
mysql> SELECT MatchID FROM Batting, Player WHERE PID=PlayerID AND
Lname='Tendulkar';
```

OR

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT
PlayerID FROM Player WHERE LName='Tendulkar');
```

```
MySQL localhost:3306 ssl d6_10 SQL > SELECT MatchID
-> FROM Batting, Player
-> WHERE PID = PlayerID
-> AND LName = 'Tendulkar';
```

MatchID
2689
2690
2691
2692
2693
2695

```
5 rows in set (0.0006 sec)
```

Q2. Find match information of all matches in which Dohni has batted.

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player;
```

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN
(SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player
WHERE LName ='Dhoni'));
```

OR

Using Join

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player WHERE
ODIMatch.MatchID=Batting.MatchID AND PlayerID=PID AND LName='Dhoni';
```

```
MySQL localhost:3306 ssl d6_10 SQL > SELECT Team1, Team2, Ground, Date
-> FROM ODIMatch, Batting, Player
-> WHERE ODIMatch.MatchID = Batting.MatchID
-> AND PlayerID = PID
-> AND LName = 'Dhoni';
```

Team1	Team2	Ground	Date
India	Sri Lanka	Colombo	2010-08-05
India	Australia	Melbourne	2010-08-10
Australia	India	Brisbane	2010-08-20
India	Sri Lanka	Melbourne	2010-08-25
India	Australia	Sydney	2010-09-05

5 rows in set (0.0005 sec)

Q3. Find match Ids of all matches in which Tendulkar batted.

mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT

PlayerID FROM Player WHERE LName='Tendulkar'); **OR**

Using Join

mysql> SELECT MatchID FROM Batting,Player WHERE

PID=PlayerID AND LName='Tendulkar';

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT MatchID
-> FROM Batting, Player
-> WHERE PID = PlayerID
-> AND LName = 'Tendulkar';
```

MatchID
2689

1 row in set (0.0015 sec)

Q4. Find Ids of all players that have both bowled and batted in the ODI match 2689

mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID
IN (SELECT PID FROM Bowling WHERE MatchID=2689);

```
MySQL localhost:3306 ssl d6_10 SQL > SELECT PID
-> FROM Batting
-> WHERE MatchID = 2689
-> AND PID IN
-> (
->     SELECT PID
->     FROM Bowling
->     WHERE MatchID = 2689
-> );
```

PID
27003
27005
27009
27010

4 rows in set (0.0013 sec)

Q5. Find Ids of all players that have either bowled or batted (or did both) in ODI match 2689

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN
(SELECT PID FROM Bowling WHERE MatchID=2689);
```

Using Union

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 UNION
(SELECT PID FROM Bowling WHERE MatchID=2689); +-----+
```

```
MySQL localhost:3306 ssl d6_10 SQL > SELECT PID
-> FROM Batting
-> WHERE MatchID = 2689
-> UNION
-> SELECT PID
-> FROM Bowling
-> WHERE MatchID = 2689;
```

PID
27001
27002
27003
27005
27009
27010

6 rows in set (0.0061 sec)

Q6. Find Ids

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT
IN (SELECT PID FROM Bowling WHERE MatchID=2689); +-----+
```

```
MySQL localhost:3306 ssl d6_10 SQL > SELECT PID
-> FROM Batting
-> WHERE MatchID = 2689
-> AND PID NOT IN
-> (
->     SELECT PID
->     FROM Bowling
->     WHERE MatchID = 2689
-> );
```

PID
27001
27002

Q7. Find match Ids where Tendulkar has played.

```
mysql> SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player
WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID; +-----+
```

```
MySQL localhost:3306 ssl d6_10 SQL > SELECT MatchID
-> FROM Batting,
-> (
->     SELECT PlayerID
->     FROM Player
->     WHERE LName = 'Tendulkar'
-> ) ST
-> WHERE PID = ST.PlayerID;
```

MatchID
2689
2690
2691
2692
2693
2695

6 rows in set (0.0036 sec)

Q8.Find match Ids of all those matches in which player 27001 bats and makes more runs than he made at every match he played at Colombo

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT
```

NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT MatchID
-> FROM Batting
-> WHERE PID = 27001
-> AND NRuns > ALL
-> (
->     SELECT NRuns
->     FROM Batting
->     WHERE PID = 27001
->     AND MatchID IN
->     (
->         SELECT MatchID
->         FROM ODIMatch
->         WHERE Ground = 'Colombo'
->     )
-> );
```

MatchID
2689

1 row in set (0.0013 sec)

Q8.Find match Ids of all those matches in which player 27001 has scored minimum runs than he made at every match he played at Brisbane

mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT MatchID
-> FROM Batting
-> WHERE PID = 27001
-> AND NRuns < ALL
-> (
->     SELECT NRuns
->     FROM Batting
->     WHERE PID = 27001
->     AND MatchID IN
->     (
->         SELECT MatchID
->         FROM ODIMatch
->         WHERE Ground = 'Brisbane'
->     )
-> );
```

MatchID
2755

1 row in set (0.0007 sec)

Q9(a). Find Ids and scores of players who scored less than 75 but more than 50 in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND
74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE
Ground='Colombo');
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID, NRuns
-> FROM Batting
-> WHERE NRuns BETWEEN 51 AND 74
-> AND MatchID IN
-> (
->     SELECT MatchID
->     FROM ODIMatch
->     WHERE Ground = 'Colombo'
-> );
```

PID	NRuns
25001	71
91001	60

2 rows in set (0.0006 sec)

Q9(b). Find Ids and scores of players who scored greater than 49 runs in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN
(SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'); +-----+-----+
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID, NRuns
-> FROM Batting
-> WHERE NRuns > 49
-> AND MatchID IN
-> (
->     SELECT MatchID
->     FROM ODIMatch
->     WHERE Ground = 'Colombo'
-> );
```

PID	NRuns
25001	71
91001	60

2 rows in set (0.0006 sec)

Q10. Find the player Ids of those players whose date of first match (FTest) is not given in the database.

```
mysql> SELECT PlayerID FROM Player WHERE FTest IS NULL;
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PlayerID
-> FROM Player
-> WHERE FTest IS NULL;
```

```
+-----+
| PlayerID |
+-----+
| 27001 |
+-----+
```

1 row in set (0.0017 sec)

Q11. Find names of the player who batted in match 2689

mysql> SELECT FName,LName FROM Player WHERE EXISTS (SELECT *
FROM Batting WHERE PlayerID=PID AND MatchID=2689); **OR**

mysql> SELECT FName,LName FROM Player,Batting WHERE PlayerID = PID AND
MatchID=2689;

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT FName, LName
-> FROM Player
-> WHERE EXISTS
-> (
->     SELECT *
->     FROM Batting
->     WHERE PlayerID = PID
->     AND MatchID = 2689
-> );
```

```
+-----+-----+
| FName | LName |
+-----+-----+
| Sachin | Tendulkar |
| Yuvraj | Singh |
| MS | Dhoni |
| Adam | Gilchrist |
| Ricky | Ponting |
| Andrew | Symonds |
| Breet | Lee |
| Praveen | Kumar |
| Harbhajan | Singh |
+-----+-----+
```

9 rows in set (0.0008 sec)

Q12. Find player Ids of player who have scored more than 30 in every ODI match that they have played.

mysql> SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT *
FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31); +-----+

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID AS PlayerID
-> FROM Batting b1
-> WHERE NOT EXISTS
-> (
->     SELECT *
->     FROM Batting b2
->     WHERE b1.PID = b2.PID
->     AND b2.NRuns < 31
-> );
```

PlayerID
24001
25001
25001
89001
91001

5 rows in set (0.0008 sec)

Q13. Find the names of players that have batted in all the ODI matches in Melbourne that are in the database.

```
SELECT FName, LName
FROM Player
WHERE NOT EXISTS
(SELECT *
FROM ODIMatch
WHERE Ground = 'Melbourne'
AND NOT EXISTS
(SELECT *
FROM Batting
WHERE PID=PlayerID
AND (Batting.MatchId=ODIMatch.MatchID));
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT FName, LName
-> FROM Player
-> WHERE NOT EXISTS
-> (
->     SELECT *
->     FROM ODIMatch
->     WHERE Ground = 'Melbourne'
->     AND NOT EXISTS
->     (
->         SELECT *
->         FROM Batting
->         WHERE PID = PlayerID
->         AND Batting.MatchID = ODIMatch.MatchID
->     )
-> );
```

Empty set (0.0010 sec)

Q14. Find the number of players that bowled in ODI match 2689

```
mysql> SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT COUNT(*) AS NBowlers
-> FROM Bowling
-> WHERE MatchID = 2689;

+-----+
| NBowlers |
+-----+
|         4 |
+-----+
1 row in set (0.0007 sec)
```

Q15. Find the average batting score of all the players that batted in the ODI Match 2689

```
mysql> SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT AVG(NRuns) AS AvgRuns_2689
-> FROM Batting
-> WHERE MatchID = 2689;

+-----+
| AvgRuns_2689 |
+-----+
|      25.2222 |
+-----+
1 row in set (0.0005 sec)
```

Q16. Find the youngest player in the database

```
mysql> SELECT FName AS Youngest_Player FROM Player
WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT FName AS Youngest_Player
-> FROM Player
-> WHERE Yborn =
-> (
->     SELECT MAX(YBorn)
->     FROM Player
-> );

+-----+
| Youngest_Player |
+-----+
| Ishant          |
+-----+
1 row in set (0.0005 sec)
```

Q17. Find the number of players in the database from each country

```
mysql> SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT Country, COUNT(*) AS NPlayers
-> FROM Player
-> GROUP BY Country;
```

Country	NPlayers
India	10
Australia	5
West Indies	1
Sri Lanka	3
Pakistan	2
Zimbabwe	1
South Africa	4
New Zealand	2
England	1

9 rows in set (0.0005 sec)

Q18. Find batting average of each player in the database. Present the PID of each Player.

```
mysql> SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID, AVG(NRuns) AS AVGR
-> FROM Batting
-> GROUP BY PID;
```

PID	AVGR
23001	19.0000
24001	42.0000
25001	53.5000
27001	4.5000
89001	91.0000
91001	60.0000
92002	1.0000
94002	17.0000
95001	1.0000
98002	2.5000
99001	7.0000
99002	2.0000

12 rows in set (0.0007 sec)

Q19. Find batting average of each player from India in the database. Present first name of each player.

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT P.FName, AVG(B.NRuns) AS AVGR
-> FROM Player P, Batting B
-> WHERE P.PlayerID = B.PID
-> AND P.Country = 'India'
-> GROUP BY P.PlayerID, P.FName;
```

FName	AVGR
Yuvraj	19.0000
MS	53.5000
Praveen	4.5000
Sachin	91.0000
Harbhajan	2.5000

5 rows in set (0.0009 sec)

Q20. Find the average score for each player when playing against Australia

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT P.PlayerID, P.FName, AVG(B.NRuns) AS AVGR
-> FROM Player P
-> JOIN Batting B
-> ON P.PlayerID = B.PID
-> JOIN ODIMatch M
-> ON B.MatchID = M.MatchID
-> WHERE M.Team1 = 'Australia'
-> OR M.Team2 = 'Australia'
-> GROUP BY P.PlayerID, P.FName;
```

PlayerID	FName	AVGR
89001	Sachin	91.0000
23001	Yuvraj	38.0000
25001	MS	36.0000
99002	Adam	2.0000
95001	Ricky	1.0000
24001	Andrew	42.0000
99001	Breet	7.0000
27001	Praveen	7.0000
98002	Harbhajan	3.0000

9 rows in set (0.0009 sec)

Q21. Find the ID of players that had a higher average score for all players when they played against Sri Lanka.

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT B.PID
-> FROM Batting B
-> JOIN ODIMatch M
-> ON B.MatchID = M.MatchID
-> WHERE M.Team1 = 'Sri Lanka'
->      OR M.Team2 = 'Sri Lanka'
-> GROUP BY B.PID
-> HAVING AVG(B.NRuns) >= ALL
-> (
->     SELECT AVG(B2.NRuns)
->     FROM Batting B2
->     JOIN ODIMatch M2
->     ON B2.MatchID = M2.MatchID
->     WHERE M2.Team1 = 'Sri Lanka'
->          OR M2.Team2 = 'Sri Lanka'
->     GROUP BY B2.PID
-> );
```

PID
25001

1 row in set (0.0009 sec)